

# VISHAL J

AI & ML Engineer | NLP | Deep Learning | Generative AI

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## PROFILE SUMMARY

AI Engineer and Data Science fresher with hands-on experience building and deploying LLM-powered applications, RAG pipelines, and deep learning systems. Proficient in prompt engineering, context injection, and fine-tuning language models (BERT, GPT, Claude API) for production-grade AI solutions. Experienced in end-to-end AI development — from data preprocessing and model training (Transformers, CNNs, LSTMs) to real-time API integration and deployment.

## TECHNICAL SKILLS

- NLP & LLMs:** HuggingFace Transformers, BERT, GPT, Claude API, RAG Pipelines, Prompt Engineering, Intent Extraction, SpaCy, NLTK
- Deep Learning:** PyTorch, TensorFlow, Keras, CNNs, LSTMs, GANs, AutoEncoders, Model Fine-tuning, Transfer Learning
- ML & Data:** Scikit-learn, Feature Engineering, EDA, Anomaly Detection, Predictive Modeling, Pandas, NumPy, SQL, Matplotlib
- Tools:** OpenCV, Git, Docker, AWS, FastAPI, Streamlit, REST APIs, WebSocket, GitHub Pages

## KEY PROJECTS

### RailAssist — LLM-Powered Chatbot with RAG Pipeline [\[GitHub\]](#)

- Built an end-to-end RAG pipeline using Claude API with NLP-based intent extraction to transform natural language queries into structured, hallucination-free responses via prompt engineering.
- Developed multi-turn conversational memory with domain-specific logic for context-aware LLM deployment.

Stack: Python, Claude API, RAG, NLP, Streamlit

### FinIntel — AI-Powered Financial Intelligence Dashboard [\[Live Demo\]](#)

- Built a real-time financial dashboard using Claude API with RAG-style prompt engineering, injecting live prices for 15+ assets into the LLM system prompt for accurate, hallucination-free responses.
- Integrated Binance WebSocket, CoinGecko API, and Alpha Vantage for live market data; deployed on GitHub Pages.

Stack: JavaScript, Claude API, Chart.js, REST APIs, WebSocket

### YOLO-MineSafe — Vision-based Fall Detection & Emergency Alert System [\[GitHub\]](#)

- Built a real-time fall detection system using fine-tuned YOLOv8; designed a temporal verification module (20-frame window) to minimize false positives and trigger automated alerts.
- Published in IJSART (April 2026), Impact Factor: 7.883.

Stack: Python, YOLOv8, OpenCV, Deep Learning

## INTERNSHIP EXPERIENCE

### Machine Learning Intern | FFuture Wise

Aug – Sep 2024

- Fine-tuned ML models using Python; performed feature extraction, data preprocessing, and model evaluation on real-world datasets to improve predictive accuracy.

### AI & ML Intern | Svasti Technology Solutions

Jan – Feb 2024

- Implemented ML models end-to-end (preprocessing → feature engineering → evaluation) and supported real-time AI workflow deployment.

### Data Analytics Intern | VDart Academy

Jul 2025

- Applied Python, SQL, and Power BI for EDA, data cleaning, and business reporting dashboards; version control via Git.

## EDUCATION

### B.Tech (Hons.) — Artificial Intelligence & Data Science

2022 – 2026

M.A.M. College of Engineering & Technology, Trichy | CGPA: 8.32 / 10.00

## RESEARCH PUBLICATION

### YOLO-MineSafe: A Vision-based Abnormal Fall Detection and Emergency Alert Framework for Isolated Mining Workers

IJSART, Vol. 12, Issue 4, April 2026 | Paper ID: IJSARTV12I4104890 | Impact Factor: 7.883

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